

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	N.Kalyan Reddy	Reg. No.:	192424232
Date:	22-08-2026	Duration:	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

Abstract

Intelligent parking is an important smart-city application because drivers may spend significant time searching for vacant spaces, contributing to congestion, fuel consumption and emissions. This literature survey reviews recent research on IoT sensing, computer vision, machine learning, edge computing, mobile applications, reservation and digital payment for smart parking. The reviewed studies show a progression from basic occupancy monitoring toward predictive, connected and user-guided parking systems. The survey also identifies an important research gap: many systems optimize one component—such as occupancy detection or prediction—without integrating reliable real-time detection, demand forecasting, route guidance, reservation, payment, security and city-level analytics in one scalable platform.

1. Industry Context

Rapid urbanization and increasing vehicle ownership have intensified parking shortages and traffic congestion in metropolitan areas. Smart parking systems combine IoT sensors, cameras, GPS, cloud computing and artificial intelligence to obtain real-time parking information and support better urban mobility. The assigned problem statement therefore focuses on an integrated platform that can detect vacant spaces, guide drivers, predict parking demand, automate reservations, support digital payments and provide occupancy analytics.

2. Problem Statement

Drivers spend considerable time searching for available parking spaces, which can increase traffic congestion, fuel consumption and carbon emissions. Conventional parking management may not provide accurate real-time availability. An AI-powered smart parking platform should combine cameras and IoT sensors for vacancy detection, mobile guidance for drivers, machine-learning-based demand prediction, automated reservations, digital payments, occupancy monitoring and utilization reports for administrators.

3. Objectives

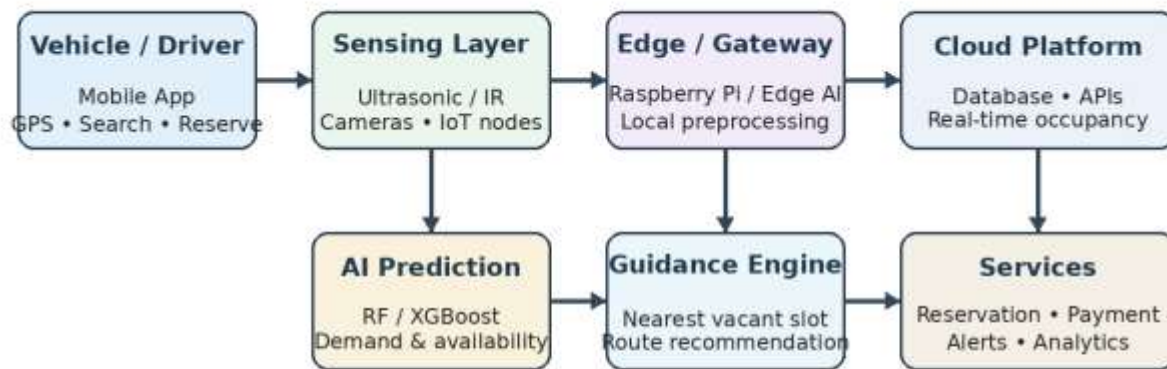
1. Detect parking availability automatically.
2. Guide drivers to vacant spaces.
3. Predict parking demand using AI.
4. Automate parking reservations.
5. Support digital payment integration.

6. Monitor parking occupancy.
7. Generate parking utilization reports.
8. Reduce urban traffic congestion.

4. Expected Outcomes

9. Reduced parking search time.
10. Lower traffic congestion.
11. Reduced fuel consumption.
12. Improved parking utilization.
13. Enhanced user convenience.

Proposed Intelligent AI-Based Smart Parking Architecture



Feedback loop: occupancy data → prediction → guidance → reservation/payment → updated occupancy

Goal: reduce search time, congestion and unnecessary fuel use while improving space utilization.

Figure 1. Proposed architecture for the Intelligent AI-Based Smart Parking and Vehicle Guidance System.

5. Literature Review

5.1 Provoost et al. (2020)

Predicting parking occupancy via machine learning in the Web of Things

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: Traffic cameras + Web of Things; neural networks and random forests; weather web services.

Major Findings: Real-world parking occupancy prediction; MSE of 7.18 at a 60-minute horizon.

Limitations: Prediction depends on sensor/camera availability and the specific study setting.

Research Gap: Need for broader, integrated real-time guidance and reservation functions [1]

5.2 Raspberry Pi-based smart campus study (2021)

An IoT Raspberry Pi-based parking management system for smart campus

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: Raspberry Pi 4, Pi camera, GPS, ultrasonic sensors, Wi-Fi and Blynk IoT dashboard.

Major Findings: Real-time monitoring and mobile visualization of outdoor parking spaces; prototype tested in a real environment.

Limitations: Campus-focused deployment and prototype-scale infrastructure.

Research Gap: Need for scalable AI prediction and wider urban deployment [2]

5.3 Elashmawi et al. (2023)

IoT Based Smart Parking System Using Ensemble Learning

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: IoT sensors, Firebase, mobile app, QR authentication, reservation, Random Forest, Linear Regression and XGBoost.

Major Findings: Integrated monitoring, reservation and automated payment; XGBoost selected as final model with score 85.7%.

Limitations: Performance and deployment are tied to the selected locations and available data.

Research Gap: Need for more robust multimodal detection, navigation and large-scale evaluation [3]

5.4 Dahiya et al. (2024)

Machine Learning-Based Prediction of Parking Space Availability in IoT-Enabled Smart Parking Management Systems

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: KNN, SVM, Random Forest, Decision Tree, Logistic Regression and Naïve Bayes; precision, recall, accuracy and F1-score.

Major Findings: Random Forest showed superior performance among the evaluated models.

Limitations: Focus is mainly on prediction; complete end-to-end user services are not the main focus.

Research Gap: Integrate prediction with guidance, reservation, payment and city analytics [4]

5.5 Hudda et al. (2024)

A WSN and vision based smart, energy efficient, scalable, and reliable parking surveillance system with optical verification at edge

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: Wireless sensor networks, ultrasonic sensing, computer vision, R-CNN/Faster R-CNN, MobileNetV2/ResNet50 and edge computing.

Major Findings: Hybrid sensing improves verification; MobileNetV2-based models were reported as at least twice as fast for processing/training in the tested setup.

Limitations: Computer-vision accuracy and performance vary with environment and model configuration.

Research Gap: Combine energy-efficient detection with predictive demand and driver guidance [5]

5.6 Knights et al. (2025)

Machine Learning Models and Mathematical Approaches for Predictive IoT Smart Parking

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: IoT parking data, machine learning and mathematical approaches for availability prediction.

Major Findings: Explores improved parking availability prediction and compares predictive approaches.

Limitations: Further real-world validation across diverse cities and operating conditions is needed.

Research Gap: Develop adaptive, city-scale predictive systems with explainability and robust deployment [6]

5.7 Alkhudhayr (2025)

Internet of things based parking slot detection and occupancy classification for smart city traffic management

Objective/Problem: Improve smart parking management through real-time occupancy information, prediction or automated parking services.

Methodology/Technology: IoT + image recognition; hybrid SSD-ShuffleNet tuned with IW-WSO and SVM-based classification.

Major Findings: Real-time parking-slot detection/classification; reported average inference time of about 50 ms per frame.

Limitations: Image-based systems still require robustness across diverse weather, lighting and layouts.

Research Gap: Combine low-cost vision with prediction, navigation and secure transactions [7]

Smart Parking and Vehicle Guidance Workflow



The workflow combines real-time sensing, AI prediction, navigation, reservation and analytics.

Figure 2. End-to-end workflow proposed from occupancy sensing to driver guidance and analytics.

6. Comparative Literature Survey

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Finding	Limitation / Gap
Provoost et al., 2020	Predict future parking occupancy	NN + Random Forest	Traffic cameras, weather services	MSE 7.18 at 60 min	Limited integrated user services
2021 smart campus study	Real-time campus parking monitoring	Raspberry Pi + camera + ultrasonic + GPS	Blynk IoT, mobile GUI	Real-time monitoring validated	Campus/prototype scope
Elashmawi et al., 2023	Integrated parking + prediction	IoT + QR + ML ensemble	Firebase, mobile app, six Cairo locations	XGBoost score 85.7%	Location/data dependent

Dahiya et al., 2024	Predict available parking space	KNN, SVM, RF, DT, LR, NB	IoT parking dataset	RF strongest overall	Prediction-centric
Hudda et al., 2024	Reliable and energy- efficient surveillance	WSN + edge vision	Ultrasonic + R- CNN/Faster R-CNN	Hybrid verification improves responsiveness	Environmental/model sensitivity
Knights et al., 2025	Predictive IoT smart parking	ML + mathematical approaches	IoT parking data	Improves predictive parking availability	More diverse real- world validation needed
Alkhudhayr, 2025	Low-cost real-time slot detection	SSD- ShuffleNet + IW-WSO + SVM	Parking images + IoT	~50 ms/frame reported	Needs broad environmental robustness

Overall comparison: the literature demonstrates strong progress in real-time occupancy sensing and machine learning prediction. The strongest direction is toward hybrid architectures that combine IoT sensing, computer vision and edge/cloud analytics. However, prediction accuracy alone does not solve the complete parking problem; the proposed system should also convert predictions into actionable guidance, reservations, secure payments and administrator analytics.

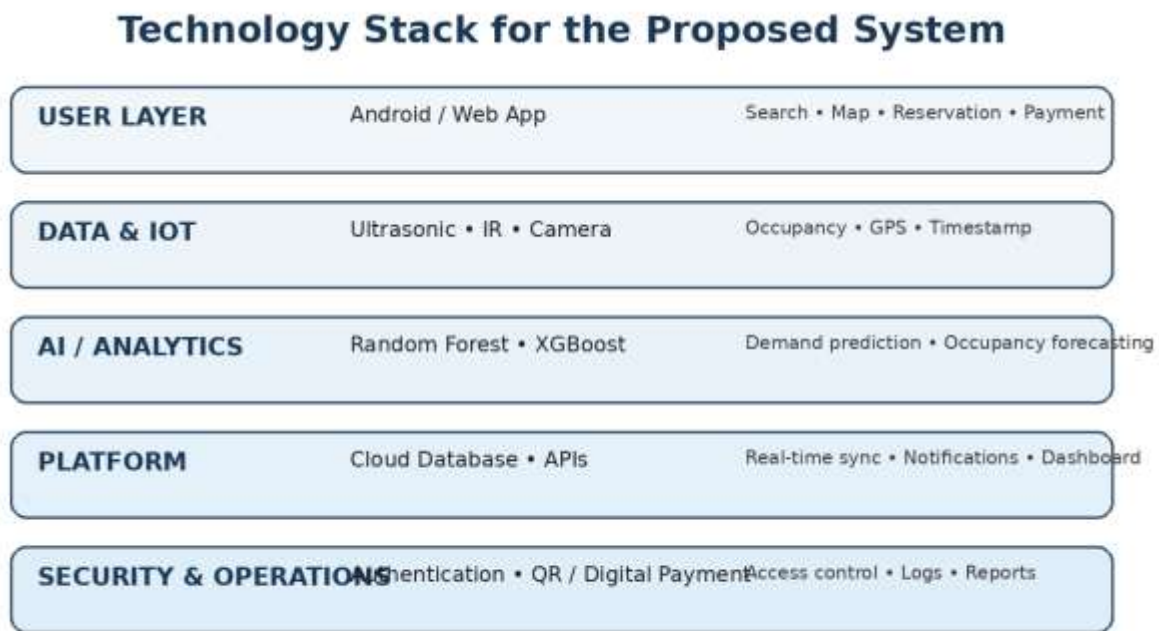


Figure 3. Technology stack connecting users, IoT sensing, AI analytics, cloud services and secure operations.

7. Critical Analysis and Research Gap

The reviewed studies show that smart parking research is moving from simple slot detection toward predictive and connected systems. Sensor-based approaches provide direct occupancy measurements, while computer-vision approaches can reduce the need for physical sensors and can cover larger visual areas. Hybrid WSN-and-vision systems improve reliability by allowing one sensing modality to verify another. Machine-learning studies show that occupancy and availability can be predicted rather than only observed.

Nevertheless, the literature reveals fragmentation. A system may provide accurate detection without route guidance; another may provide prediction without reservation; another may provide reservation and payment but remain limited to a small prototype or specific locations. Environmental conditions, network connectivity, sensor reliability, privacy, data quality, model drift and scalability also remain practical challenges.

The major research gap is therefore an end-to-end, scalable and adaptive platform that jointly performs (1) multimodal real-time vacancy detection, (2) short-term parking-demand prediction, (3) personalized vacant-slot guidance, (4) reservation and digital payment, (5) occupancy/utilization analytics, and (6)

secure and reliable data management. Such integration can turn parking prediction into a complete vehicle-guidance service rather than an isolated classification task.

8. Proposed Improvement / Future Research Directions

14. Develop a hybrid sensor-and-camera architecture with automatic sensor verification and fallback mechanisms.
15. Use adaptive machine-learning models that account for time, day, weather, events and historical occupancy.
16. Integrate GPS-based route guidance so that the selected parking slot minimizes both parking distance and expected arrival time.
17. Support advance reservation, no-show handling and secure digital payment in the same application.
18. Use edge computing for latency-sensitive occupancy detection and cloud computing for historical analytics and model training.
19. Introduce privacy-preserving methods for camera data and secure authentication for users and administrators.
20. Evaluate the complete system using metrics such as detection accuracy, prediction error, inference latency, search-time reduction, utilization and user satisfaction.
21. Test across multiple parking layouts and environmental conditions to improve generalization and scalability.

9. Conclusion

The literature survey indicates that AI, IoT, computer vision, edge computing and cloud platforms provide a strong technical foundation for intelligent parking and vehicle guidance. Existing studies demonstrate effective occupancy detection and prediction, while some systems additionally support reservations, QR authentication and automated payment. The proposed Intelligent AI-Based Smart Parking and Vehicle Guidance System builds on these directions by emphasizing an integrated workflow from sensing and prediction to guidance, reservation, payment and utilization analytics. The key research opportunity is to validate such an integrated platform at realistic urban scale while maintaining accuracy, low latency, reliability, privacy and user convenience.

10. References

- [1] J. C. Provoost, A. Kamilaris, L. J. J. Wismans, S. J. van der Drift, and M. van Keulen, "Predicting parking occupancy via machine learning in the Web of Things," *Internet of Things*, vol. 12, 100301, 2020. DOI: 10.1016/j.iot.2020.100301.
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- [6] V. Knights, O. Petrovska, J. Bunevska-Talevska, and M. Prchkovska, "Machine Learning Models and Mathematical Approaches for Predictive IoT Smart Parking," *Sensors*, vol. 25, no. 7, 2065, 2025. DOI: 10.3390/s25072065.
- [7] H. Alkhudhayr, "Internet of things based parking slot detection and occupancy classification for smart city traffic management," *Engineering Applications of Artificial Intelligence*, vol. 152, 110802, 2025. DOI: 10.1016/j.engappai.2025.110802.